Covid-19 and Help-Seeking Behavior for Intimate Partner Violence Victims

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Abstract:

This study seeks to adress two key questions about the effect of the quarantine policies on domestic violence reporting: the role of third party vs victims' reporting of IPV through hotline calls; the extent to which reported domestic violence translated into protection given to victims. In order to address the first question, we focus on the effect of the introduction of the first lockdown in Spain on the reported cases of domestic violence by victims and third parties by exploiting province-level and monthly data on the number of calls to the dedicated emergency hotline for IPV victims between 2013 and 2020. We find that the introduction of the first lockdown was responsible for an average increase in total calls of 48%, which represents an additional 26,000 calls compared to previous years. Importantly, we also show that help-seeking behavior is at the origin of 70% of all reported calls. With respect to the second question, we provide evidence of a substitution effect between calls to the emergency hotline and reporting to the police during the confinement, which led a drop in the level of protection of IPV victims. Despite the large increase in reported IPV during the pandemic, we estimate that the introduction of lockdown caused a loss of 3,200 protection orders to victims that would, in the absence of the pandemic, have received it.

Keywords: intimate partner violence, COVID-19, help-seeking behavior

JEL: J12, J16, J18, I12,

1 Introduction

The mobility restrictions and quarantine policies that have been implemented worldwide in the context of the COVID-19 outbreak have had large and unintended consequences on domestic violence. Social distancing measures, coupled with the large human and socioeconomic losses, have increased the risks of abuse faced by women and children at home, in a context where traditional channels of detection and protection of victims were no longer – or much less – accessible.

In this paper we seek to contribute to the understanding of the effect of mobility restriction policies on domestic violence reporting and incidence, thereby contributing to a growing literature (Peterman *et al.*, 2020; Arenas-Arroyo, Fernandez-Kranz and Nollenberger, 2020; Beland *et al.*, 2020; Campedelli, Aziani and Favarin, 2020; Kumar and Nayar, 2020; Leslie and Wilson, 2020; Piquero *et al.*, 2020;Miller, Segal and Spencer, 2020; Perez-Vincent *et al.*, 2020; Ravindran and Shah, 2020, to quote a few)[1]. The majority of these previous studies have found a significant increase in *reported* domestic violence during lockdown through emergency hotlines (Leslie and Wilson, 2020; Miller, Segal and Spencer, 2020; Perez-Vincent *et al.*, 2020; Ravindran and Shah, 2020) with disparities of estimated results depending on the case studied (cities vs. country), the type of data used (hotline data vs. police data), and the time frame considered (mostly short term).

This study seeks to adress two key questions which have not yet (or only partially) been adressed in the current literature. First, the role of third party reporting in the documented increase in calls to emergency hotlines in a context where neighbors were more likely to become aware of domestic violence while being at home. Addressing this question is key to properly interpret the changes in IPV reporting: does the documented increase in measured IPV steam from a mere reporting effect by third parties? Did IPV victims change their reporting strategy in the context of the quarantine? To our knowledge, an exception to this shortcoming in the literature is the paper by Ivandic *et al.* (Ivandic, Kirchmaier and Linton, 2021), in which they find that the increase in police calls related to domestic violence following lockdown were entirely driven by third party calls, rather than victims.

[1] See also the list updated by Dr Amber Peterman for a more exhaustive list of papers investigating the number of studies measuring the effect of COVID-19 on violence against women and children, accessible here: https://drive.google.com/file/d/1w28pmDt-lOIcI9cvaUdAMVVDzftOiI1-/view

Secondly, our paper also seeks to contribute to the understanding of the relationship between different channels of detection of IPV victims, and more specifically on the possible substitution between reporting through calls, and reporting to the police. As reporting to the police is a necessary step for judicial protection, understanding the importance taken by reporting through calls in the overall detection system of IPV victims is crucial from a public policy point of view.

We focus on Spain– a country which was intensively hit by both the pandemic and mobility restriction policies. Lockdown was implemented on March 15th in all regions under very strict conditions and was relaxed progressively from early June. After a rather relaxed summer, curfew was established on the 20th October as a response to the subsequent waves of the pandemic, and was not relaxed until April 2021. We explore the effect of the introduction of the first lockdown on the reported cases of domestic violence by victims and third parties by exploiting province-level[2] and monthly data on the number of calls to the dedicated emergency hotline for IPV victims between 2013 and 2020. Using a difference-in-differences strategy (with 2020 being our treated year, and March-December being our "Post" period for every year), we find that the introduction of the first lockdown was responsible for an average increase in total calls of 48%, which represents an additional 26,000 calls compared to previous years. Contrary to previous findings, we find that the implementation of lockdown caused an increase in both third-party reporting and victims' reporting, help-seeking behavior being the main channel of reporting (70% of all calls). Furthermore, we provide evidence of a substitution effect between calls to the emergency hotline and reporting to the police during the confinement, and of a significant decrease in detection and protection of victims by both the police forces and the judicial administration over the 9 months period following the pandemic outbreak. Despite the large increase in reported IPV during the pandemic, we estimate that the introduction of lockdown caused a loss of 3,166 protection orders to victims that would, in the absence of the pandemic, have received it.

Our findings have important policy implications. We provide evidence that the emergency hotline appeared to be a primary channel of help-seeking behavior for women during lockdown but also beyond, while traditional channels of protection (the police) became less accessible/safe for women. As such, we believe that emergency hotlines should be given a central role in the overall detection and protection system of IPV victims, and foster a more direct and systematic communication with police forces, women's shelters and/or health workers. A direct consequence of this substitution in reporting is the decrease in judicial protection given to women, which can

[2] There are 17 regions (Autonomous Communities), 50 provinces and 2 autonomous cities (Ceuta and Melilla)

only be acquired through former complaints to the police forces: significant efforts should be engaged to detect women facing the highest risks of abuse, and to fasten judicial administrative processes related to protection orders' decisions. We also show that while help-seeking behavior is the main channel of reporting, third party reporting also represented a key channel in the immediate aftermath of the implementation of the lockdown. As such, we argue that awareness campaigns directed to the general population aiming to raise concerns and empathy towards IPV victims might be an efficient mechanism of increasing detection of IPV victims through third parties.

The first section of this paper is dedicated to the analysis of the effect of lockdown introduction on changes in calls to the emergency hotline, according to the type of caller. In the next section, we investigate differences across provinces in terms of lockdown introduction effect, focusing particularly on differences in mobility reduction during confinement. Finally, we exploit police and judicial data on IPV related complaints and protection orders to explore the relationship between these different reporting channels and protection of IPV victims.

2 The effect of the national-level quarantine measures on victim and third-party reporting

In this first section, we explore the effect of the implementation of the national-level quarantine measures on victims' and third-party reporting, both on average and overtime between March 2020 and March 2021. National lockdown was implemented on March 15th in all regions under very strict conditions and was relaxed progressively from early June. After a "relaxed summer" (mobility between provinces and countries lifted, cultural events allowed, bars and restaurants opened, etc.), curfew was established on the 20th October as a response to the subsequent waves of the pandemic, and was not relaxed until April 2021[3].

2.1 Data: Calls to the emergency hotline 016

We use as measure of reported domestic violence (DV) the number of calls to the national emergency hotline 016 (Government Office on Gender-based violence), by province, year and month and according to the type of caller: the victim, a family member of the victim, or outside of

[3] See section **A.a** for further details on the chronology of social distancing measures in Spain between March 2020 and March 2021)

the family reporting. While data is available from 2008 to 2020[4], we restrict our analysis to the period 2013-2020 as years from 2008 to 2012 were affected by large external shocks (economic crisis, major labor reforms) likely to affect the comparability of 2020 with those years.

Our final dataset consists of 4,992 cells defined at the province x month x year level between January 2013 and March 2021. We define 3 categories of outcomes depending on the type of caller (all callers, victims, third parties) and construct for each category two outcomes of interest: the number of calls per 100,000 inhabitants and the log number of calls.

- $CallsHab^{tot}_{p,y,m}$ is the total number of calls per 100,000 inhabitants and $Log(Calls^{tot}_{p,y,m})$ the log number of calls in province p, year y and month m.
- $CallsHab^{V}_{p,y,m}$ is the number of calls *made by victims themselves* per 100,000 inhabitants and $Log(Calls^{V}_{p,y,m})$ the log number in province p, year y and month m.
- $CallsHab^{Th}{}_{p,y,m}$ is the number of calls *made by third parties themselves* per 100,000 inhabitants and $Log(Calls^{Th}_{p,y,m})$ the log number in province p, year y and month m.

The yearly number of calls made to the emergency hotline in Spain between 2013 and 2020 ranges from 65,000 to 85,000, with the highest number of calls registered in 2015 (**Fig A 1**): this is equivalent to yearly calls per 100,000 inhabitants ranging from 138 to 180. The pandemic outbreak early 2020 coincides with a sharp increase in both the number of calls made by victims and third parties, thereby breaking the overall decreasing trend in calls to the hotline registered since 2015. Over the considered years, the share of third-party calls remained stable around 30%.

2.2 Identification strategy

We exploit province-monthly level information on the number of calls to 016 between 2013 and 2020 and formally quantify the effect of the pandemic outbreak on reported domestic violence using a difference-in-differences framework, where our "treated" year is the post-pandemic outbreak year (2020) and our "control" years are the pre-pandemic years from 2013 to 2019. The pre-policy (lockdown) period that we consider for each control and treatment year are the months of January and February; the post-policy months are March to December. **Fig A 2** in the appendix displays changes in calls relative to February for both the treatment and control group. We estimate average effects using a difference in differences specification and investigate the dynamics of this effect in the short- and medium-term using event study models.

[4] We do not include in our analyses data from 2021, which is only available for the months of January and February.

Estimation of the average effect of quarantine implementation between March and December 2020

Our econometric specification for the estimation of the average effect of lockdown introduction on domestic violence reporting is the following:

$$\textbf{\textit{Eq 1}}.\ CallsHab^{j}_{p,y,m} = \alpha + \beta_0 D_{y=2020} + \beta_1 D_{m\geq 3} + \beta_{did} D_{y=2020} D_{m\geq 3} + \gamma_p + \gamma_y + \gamma_m + \epsilon_{p,y,m}$$

With *j= {Tot; V; Th}* the indicator for total, victim, or third-party calls per 100,000 inhabitants. $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March[5], and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years. We include province-level fixed effects to account for time-invariant characteristics likely to affect IPV reporting through the hotline (demographic characteristics, existence of local hotlines, population density, etc.); year fixed-effects to account for common time trends across provinces (*e.g* higher level of calls during economic recessions); month fixed effects to control for seasonality changes in IPV reporting across provinces and year[6].

We run the same regression on the log of calls controlling for population size ($Pop_{p,y}$) which allows us to interpret regression coefficients in terms of percentage change in calls.

$$\textbf{\textit{Eq 2}}.\ Log\left(Calls^{j}_{p,y,m}\right) = \alpha + \beta_0 D_{y=2020} + \beta_1 D_{m\geq 3} + \beta_{did} D_{y=2020} D_{m\geq 3} + \gamma_p + \gamma_y + \gamma_m + Pop_{p,y} + \epsilon_{p,y,m}$$

In alternative specifications, we add time-varying additional controls likely to affect IPV reporting in each province: total unemployment rate, male and female unemployment rate, population density.

Estimation of the dynamic effect of quarantine implementation between March and December 2020

After estimating the average effect of quarantine implementation between March and December 2020, we explore the timing of the effect on reported DV in the short- and medium-term by

[5] We consider an intensity of treatment of 0.5 for the month of March since lockdown was implemented on the 15th.

[6] $D_{m\geq 3}$ partially controls for seasonality by isolating, respectively, January-March from April-December. The inclusion of month fixed effects allows to control for seasonality within each of the subgroups defined by $D_{m\geq 3}$.

running event study models on the log number of calls to 016 on all seven years of reporting data (2013-2020):

$$\textbf{\textit{Eq 3.}}\ Log(Calls^{j}_{p,y,m}) = \alpha + \beta_0 D_{y=2020} + \sum_{i=-2}^{9} D_{m=3+i} + \sum_{i=-2}^{9} \eta_i D_{m=3+i} D_{y=2020} + \gamma_p + \gamma_y + Pop_{p,y} + \epsilon_{p,y,m}$$

We specify dummies ($D_{m=3+i}$) for the two months of lag before lockdown implementation, and 9 weeks of leads up to December to capture the long-term effect of the quarantine on DV reporting. For each of these dummies, η_i captures the additional effect of the pandemic outbreak on the considered month in 2020, relative to control years (2013-2019).

2.3 Results

Regression results for our three variables of interest (total calls, calls by victims, calls by third parties) expressed per 100,000 inhabitants and log-transformed are presented below:

Table 1. Impact of lockdown introduction on monthly domestic violence reporting through emergency hotline – difference-in-differences estimates.

	(1) Log of calls	(2) Log of calls by victims	(3) Log of calls by third parties	(4) Total calls per 100,000	(5) Calls by victims by 100,000	(6) Calls by third parties by 100,000
$\boldsymbol{D_{y=2020}D_{m\geq 3}}$	**0.390*** **	**0.380*** **	**0.414*** **	**4.446*** **	**2.983*** **	**1.463*** **
	(0.026)	(0.030)	(0.035)	(0.347)	(0.270)	(0.116)
$\boldsymbol{D_{m\geq 3}}$	0.040+	0.034	0.027	0.358+	0.201	0.157+
	(0.023)	(0.023)	(0.029)	(0.187)	(0.133)	(0.086)
$\boldsymbol{D_{y=2020}}$	0.039	0.065*	-0.025	0.447+	0.490*	-0.043
	(0.025)	(0.030)	(0.030)	(0.236)	(0.206)	(0.070)
Constant	3.838***	3.438***	2.712***	10.470***	8.260***	2.210*
	(0.024)	(0.028)	(0.028)	(2.879)	(1.918)	(1.024)
Observations	4990	4984	4949	4992	4992	4992
R^2	0.214	0.176	0.131	0.249	0.209	0.159
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March[7], and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years.
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

[7] We consider an intensity of treatment of 0.5 for the month of March since lockdown was implemented on the 15th.

We estimate that the introduction of the first quarantine policies following the pandemic outbreak is responsible for an average increase in IPV reporting of 47.7%[8] between March 2020 and December 2020 compared to previous years over the same period. this corresponds to an additional +26,345[9] compared to 2019 levels (**Table A 2** in appendix), and is equivalent to an additional +4,5 calls per 100,000 inhabitants each month[10] (Specification 4).

Estimation results from specifications 4 to 6 allow us to disentangle precisely to what extent help-seeking behavior of IPV victims on the one hand, and reporting by third parties on the other hand, contributed to the increase in reported DV that we document. We find that 67% of the increase in reported DV between March and December 2020 by reporting by victims' themselves against 33% for third party calls[11], and that the composition of reported DV in terms of who is reporting the violence changes only very slightly after lockdown (38% increase for victims' reporting against 41% for third parties). Our results are opposite to Ivandic, Kirchmaier and Linton's findings that the increase in reported DV hotline calls in the Greater London area during quarantine steamed exclusively from third parties reporting. While the latter provide indirect evidence that this does not reflect a pure reporting effect[12], we are able to directly rule out this hypothesis in the context of an entire country (Spain), over a 9-month period.

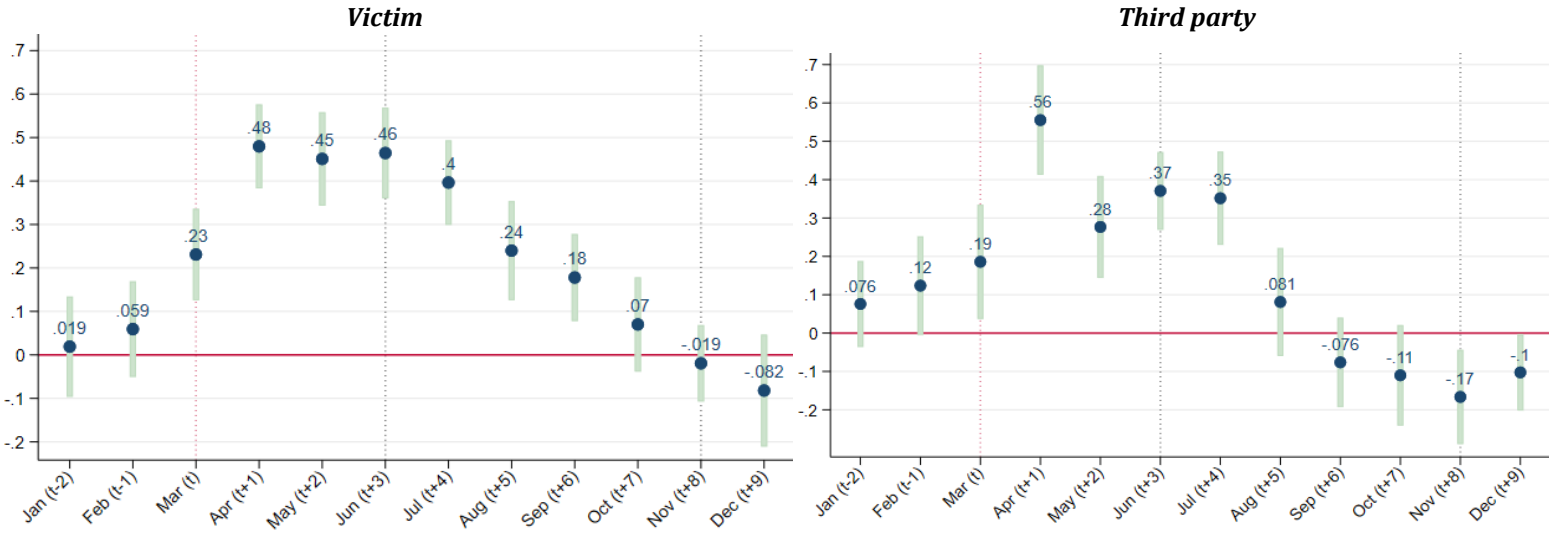
Results are robust to alternative specifications in which we control for population density and unemployment rate (See **Table A 3 Table A 4** and **Table A 5** in the appendix)[13].

We next proceed to investigate the dynamics of the effect of lockdown overtime and according to the type of caller. Event study plots of the effect of quarantine implementation in March 2020 provide suggestive evidence of the existence of common pre-trends in the first two months between the treated and control years for all calls (**Fig A 3**), and for calls by victims and third parties (**Figure 1**). The first red dotted line indicates the month of implementation of the national quarantine measures in Spain (15th of March), which should be considered as a partially treated month. The second grey dotted line marks the beginning of a more "relaxed" period in terms of

[8] Exp(0.39)-1
[9] 0.477x55,233
[10] the number of calls per 100,000 registered during the same period in 2019 was equal to 99.8, see Table A 2.
[11] To obtain the share of additional DV reporting due to help-seeking behavior, we divide the estimated additional calls by victims (specification 5) by the additional total calls.
[12] Using information on third party reporting during working hours and evening hours
[13] We note that coefficient estimates regarding the effect of unemployment rate on reported IPV should be interpreted with caution given the endogeneity of this variable in our model. The stability of our results to the introduction of this variable shows, however, that changes in reported IPV are driven by other channels than the sole economic instability generated after the pandemic.

mobility restriction[14] which lasted over the summer months up until mid-October. The last dotted grey line marks the beginning of a second wave of restriction, which included national curfew at 10.00pm and the closure of bars and restaurants.

Figure 1. Impact of lockdown introduction on domestic violence reporting through emergency hotline, according to the type of caller. Event study estimates: log models



While results presented in the previous section show that lockdown introduction had the same effect on the average increase of calls by victims and third parties in terms of percentage change between March and December 2020, we see here that the *dynamics* of this effect overtime is different according to the type of caller. We observe that the strongest effect of the lockdown on third party reporting was registered in April 2020, which is the first fully treated month (75% increase[15]). This effect loses in magnitude in the two following months of strict quarantine (May and Early June), and becomes non statistically different from 0 from August onwards. On the other hand, the increase in victims' reporting remained at 58%[16] in April, May, and June, and only started to (slowly) lose in magnitude in the "relaxed summer" phase to become non-statistically different from 0 in November 2020 (against August 2020 for third party reporting).

Event study estimates for changes in help-seeking behavior indicate that the effect of quarantine introduction had persisting effect beyond the duration of confinement (which is indicated in the figures by the red dotted line and the grey dotted lone). The persistent effect of quarantine

[14] A chronology of the main phases of the quarantine and social distancing measures between March 2020 and May 2021 is given in the appendix, section **A.a.**
[15] Exp(0.56)-1
[16] Exp(0.46)-1

introduction on the help-seeking behavior of IPV victims could be explained by an inertia of violence, whereby violent conflict in the household is more likely to occur once previous violent events have already taken place, even when the conditions that gave rise to the previous violent aggressions are no longer the same. This could also reflect a "delay" in reporting, whereby victims who were not able to contact the hotline while being all the time with the violent partner report the domestic abuse after confinement was lifted.

This spike in third party reporting immediately after lockdown could steam from an awareness effect, whereby neighbors - being always at home - could now witness domestic violence episodes that they would otherwise not have noticed. This could also reflect a strong increase in domestic violence itself immediately after lockdown. While we are not able to precisely identify the drivers of these differences, our results are nonetheless insightful as they provide evidence that third party reporting might be very sensitive to large shocks such as the quarantine implementation, which makes it a key actor in the overall system of detection and protection of IPV victims.

3 Heterogeneity analyses across provinces: the role of *effective* mobility restriction and pre-pandemic characteristics

To shed light on the possible mechanisms underlying reported DV's response to quarantine implementation, we ran heterogeneity analyses for subgroups of provinces likely to experience significant differences in levels of IPV reporting during the pandemic. We explore three categories of factors: (1) mobility reduction during confinement; (2) pre-pandemic level IPV incidence, which we measure using quarterly police data, protection orders, and calls to the emergency hotline (3) pre-pandemic demographic and labor characteristics, measured by population and share of essential workers.

3.1 Data

Mobility reduction during confinement

One key hypothesis of this paper and in the overall literature on the effect of the pandemic on the incidence of IPV is that mobility restrictions, and particularly unprecedented, forced coexistence, increased the risks faced by women and children to suffer violence from men at home. In order to quantify the extent of this relationship, we use information from the COVID-19 mobility database of the National Institute of Statistics (INE) on the share of the mobile population at the province level, between March 2020 and June 2020 and during a reference week in November 2019 (we present in further details the mobility data from the INE and how we measure mobility reduction during confinement in the annexes, section **A.b**).

Fig A 4 in the appendix displays the average decrease in the share of the mobile population in each Spanish region (CC.AA), compared to pre-pandemic (November 2019) levels. While formal mobility was equally restricted on the entire Spanish territory during the initial phase of the lockdown measures, *effective* mobility reduction differed across Spanish autonomous communities.

Mobility restriction differences during confinement could steam from pre-pandemic province characteristics (such as demographics, labor market structure, etc.), from differences in the evolution of the pandemic across the Spanish territory– with mobility being more restricted in the most hit provinces, either because of a stronger enforcement of the confinement measures by the local police or because of a stricter quarantine compliance by the population due to the fear of being infected/spreading the virus – or because of a higher unemployment rate as a consequence of the pandemic. Using an OLS regression of the monthly decrease in mobility on population density, COVID-19 incidence, and unemployment during confinement, we find that these variables do not explain significantly differences in mobility reduction across provinces. We therefore consider differences in mobility reduction during confinement as a separate potential factor of IPV reporting, beyond population density and labor market characteristics.

We define high and low mobility reduction provinces using the median value of the average decrease in mobility during confinement, calculated over all provinces. Formally, high mobility reduction provinces are those whose average decrease in mobility during confinement (compared to November 2019) is below (higher in absolute value) the median; low mobility reduction provinces are those whose average decrease in mobility compared to November 2019 is above or equal to (below in absolute value) the median. Descriptive statistics on the average mobility reduction in Spain (**Table A 1**) and evolution of the number of calls in high and low mobility reduction provinces (**Fig A 5**) are provided in the appendix,**.**

Pre-pandemic levels of IPV incidence

We now consider whether social distancing increased violence among provinces with the highest IPV incidence before the pandemic outbreak (intensive margin) or caused an increase in violence in provinces with no or low history of domestic abuse (extensive margin). Differences in pre-pandemic IPV incidence are measured using police data at the province-quarter of year level on the number of complaints related to IPV filed by either the victim or the police. We also consider protection orders at the province and quarter of year level. Plots of the evolution of IPV police reports and protection orders per 100,000 inhabitants are presented in the appendix, **Fig A 6.**

For each measure of IPV incidence per 100,000 inhabitants, the threshold used to differentiate provinces into either the high or low subgroup is the median of the province average between 2015 and 2018.

Pre-pandemic demographic and labor market structure

Only workers from essential sectors were allowed to commute during the strict quarantine period. We could therefore expect the share of essential workers[17] to affect mobility and employment during quarantine, which in turn is likely to affect IPV incidence (less time spent at home, less economic uncertainty) and reporting (neighbors less likely to witness the abuse). Population density, on the other hand, is likely to affect third party reporting through the probability of hearing the abuse.

We investigate the role of these pre-pandemic labor market and demographic characteristics on changes in reported IPV by differentiating provinces between high and low essential workers / population density provinces. Our measure of provinces' share of essential workers is taken from the INE, as defined by Royal Decree 10/2020 from the 29th of March. We take as reference value for the pre-pandemic density of population per provinces as of the 1st of January 2019 (INE). We define as high density provinces those whose density in 2019 belongs to the top 25th quantile of the distribution of population density so as to identify specifically highly population areas. High vs. low share of essential workers is defined based on the median value of the share of essential workers.

3.2 Identification strategy

We estimate the additional effect of the quarantine on reported IPV based on provinces' characteristics using data from 2013 to 2020, with a model of the form:

$$\textbf{\textit{Eq 4}}.\ Log(Calls^{i}_{g,m,p,y}) = \alpha_0 + \beta_1 D_{m\geq 3} + \beta_2 D_{y=2020} + \beta_{did} D_{m\geq 3} D_{y=2020} + \delta_{ddd} D_g D_{m\geq 3} D_{y=2020} + \gamma_p + \gamma_m + \gamma_y + \epsilon_{w,p,y}$$

g is the criteria that we use to distinguish our provinces (pre-pandemic IPV incidence, density, etc.). $Log(Calls^{i}_{g,m,p,y})$ is the log number of reported IPV episodes for the type of caller i (all callers, victims or third parties) when considering differences between provinces related to g. D_g is a dummy indicator dividing the provinces into the high and low subgroups: the group of interest

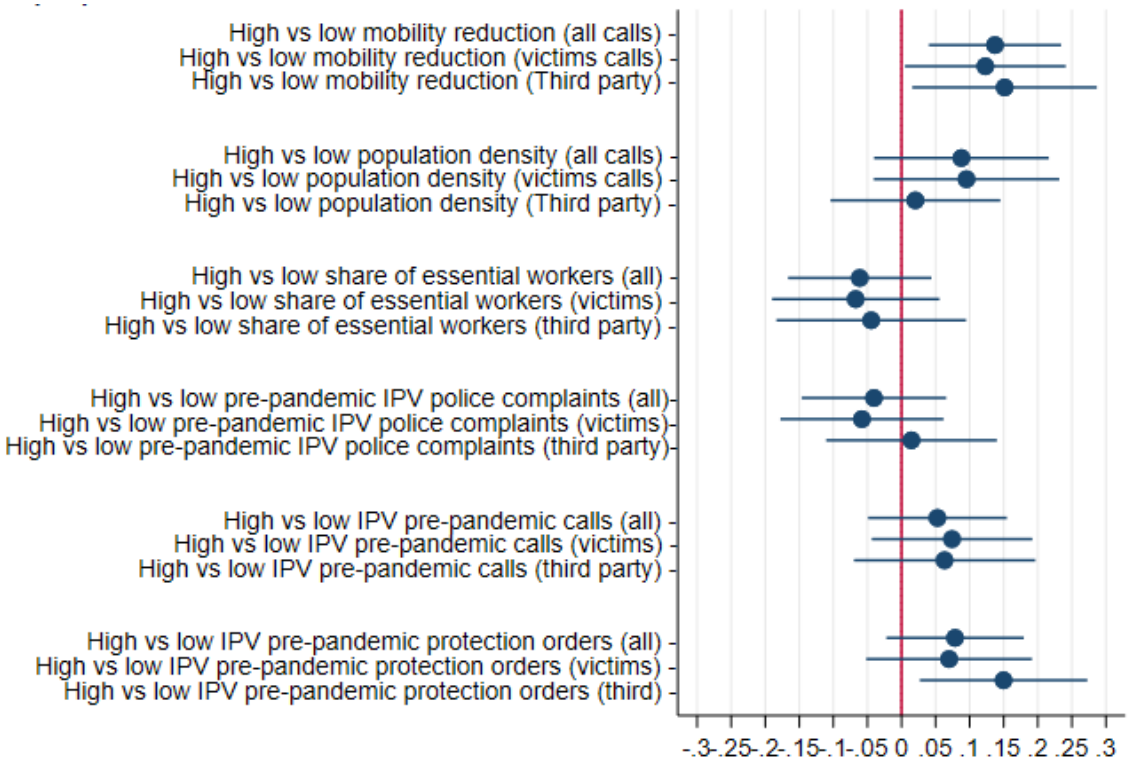
[17] As defined by Royal Decree 10/2020 based on employment composition by provinces on the 29th of March 2020.

D_g=1 (e.g. provinces with high mobility reduction) and the comparison group defined by D_g=0 (e.g. provinces with low mobility reduction). β_{did} measures the average effect of the lockdown on changes in reported DV. δ_{ddd} measures the *additional* effect of lockdown for provinces of type D_g=1, compared to the rest of provinces. The total effect of the introduction of lockdown on the subgroup D_g=1 is equal to β_{did}+δ_{ddd}. To account for the fact that mobility reduction during confinement is likely to be partially affected by population density, we control for this variable in the heterogeneity analysis according to mobility reduction[18].

3.3 Results

Results of the heterogeneity analyses are presented in the figure below, which plots the value of the δ_{ddd} coefficients (**Eq 4**)

Table 2. Additional effect of the quarantine measure according to mobility reduction and pre-pandemic characteristics, in % change (log models)



Contrary to previous findings in the literature, we do not find that the increase in IPV reporting is driven by provinces with the lowest IPV incidence before the pandemic (Leslie and Wilson, 2020).

[18] The share of essential workers, which also likely affected differences in mobility across provinces, is accounted for by province fixed effects in the regression.

While coefficients are not statistically different from 0, we see on the contrary that provinces with higher pre-pandemic protection orders per 100,000 inhabitants exhibit a higher increase in reported IPV after lockdown, in particular for third party reporting (15% higher). As protection orders represent one of the most advanced forms of victims' protection and are usually dictated for extreme situations of repeated domestic violence, we interpret this result as increases in IPV reporting being stronger in provinces with a historically higher incidence of "extreme" abuse (intensive margin).

We find that provinces with the highest mobility reduction during confinement experienced the strongest increase in the number of calls (+16%) compared to low mobility reduction provinces. To verify that the mobility reduction effect that we estimate does not measure the same effect as pre-pandemic IPV incidence (i.e., that provinces with higher mobility reduction are not the same as those with higher IPV incidence), we measure the correlation between average mobility reduction during confinement and pre-pandemic protection orders per 100,000. We find a positive but weak correlation[19] between mobility and pre-pandemic protection orders, which suggests that pre-pandemic levels of protection orders and mobility are distinct factors of increase in IPV reporting.

Overall, we find that mobility reduction and previous history of IPV incidence are the main driver of differences in IPV reporting across provinces. This suggests, in turn, that violence is highly sensitive to time spent at home, and that violence might have increased at the intensive margin in the pandemic aftermath.

4 Substitution between channels of reporting and victims' protection

How much of the increase in reported IPV that we documented in the previous sections reflects a substitution effect between hotlines and traditional channels of detection of IPV? Did victims adapt their help-seeking behavior to the constraint of the pandemic, during confinement but also beyond? If so, what are the consequences of this substitution on the overall protection of victims?

The objective of the last section of our paper is twofold: (i) to discuss the extent to which the sharp increase in reporting IPV through hotline calls by victims steams from a substitution effect between hotline reporting and reporting to the police (ii) estimate the loss in victims' protection caused by the pandemic outbreak, as measured by police interventions and protection orders.

[19] Pearson correlation coefficient of 0.1567

4.1 Methodology

In order to do so, we exploit quarterly-province level data on the number of police complaints for IPV filed either by the victims, or by the police between 2013 and 2020. We also exploit information from the Ministry of Justice on the quarterly number of protection orders per province between 2013 and 2020.

We formally investigate the effect of the pandemic on the number of police reports filed for IPV and protection orders using the following difference-in-difference equations:

$$\textbf{\textit{Eq 5}}.\ CrimeHab^{j}_{p,y,q} = \alpha + \beta_0 D_{y=2020} + D_{q\geq 2} + \beta_{did} D_{y=2020} D_{q\geq 2} + \gamma_p + \gamma_y + \gamma_q + Pop_{y,p} + \epsilon_{p,y,q}$$

$$\textbf{\textit{Eq 6}}.\ Log(Crime^{j}_{p,y,q}) = \alpha + \beta_0 D_{y=2020} + D_{q\geq 2} + \beta_{did} D_{y=2020} D_{q\geq 2} + \gamma_p + \gamma_y + \gamma_q + \epsilon_{p,y,q}$$

$Crime^{j}_{p,y,q}$ is the number of police complaints for IPV (total, filed by the victim or due to direct police intervention) or the number of protection orders in province *p,* quarter of year *q,* and year *y.* We explore the effect of the lockdown introduction on the outcomes expressed per 100,000 inhabitants (**Eq 5**), and log transformed (**Eq 6**). In both models, $D_{q\geq 2}$ is a dummy variable which is equal to 1 from the 2nd quarter to the last quarter of each year. We run the same models for the number of protection orders per 100,000 inhabitants and expressed in log.

4.2 Results

Regression results for the three category indicators measured at the quarterly level are presented **Table A 7** in the appendix. We find that police reports filed for IPV did not increase significantly following quarantine measures compared to previous years (specifications 1-2 and 3-4). This result is robust to running the model on police reports per 100,000 inhabitants, or on the log number of police reports. We find, on the other hand, that the number of protection orders decreased significantly after lockdown implementation in 2020 compared to what would have been expected in the absence of COVID-19. More specifically, we find that the quarantine measures introduced at the very end of the 1st quarter of 2020 caused a decrease of -6.75 protection orders per 100,000 inhabitants[20] over the rest of the year, itself representing an "loss" of 3,166 protection orders to victims that would, in the absence of the pandemic, have received it[21].

[20] Quarterly effect of lockdown introduction on protection orders, multiplied by three (-2.249x3)
[21] This back of the envelope calculation was done by multiplying the yearly effect (-6.75) by the total Spanish population in 2020 (46 937 056), divided by 100 000.

Given the large increase in IPV reporting by both victims and third parties that we document in this paper and previous evidence on the effect of the quarantine on reported and unreported domestic violence in Spain (Arenas-Arroyo, Fernandez-Kranz and Nollenberger, 2020), it is very unlikely that these results would be explained by a decrease in IPV prevalence. Rather, we argue that the pandemic outbreak and subsequent lockdown measures caused a loss of detection and protection of IPV by the police forces, and a loss of protection granted by the judicial administration.

This loss in detection could steam from a substitution to alternative forms of reporting by victims, as documented in this paper, who might be less willing (or less able) to report to the police in the context of the quarantine measures. As judicial protection of victims cannot be issued without proper complaints filed to or by the police, this substitution might explain the decrease in protection orders that we document. At the same time, part of the reduction in protection orders could come from slower judicial procedures in the context of the pandemic, during which judges and prosecutors had to work from home (as opposed from police forces).

Overall, our results point to the crucial need to detect the most vulnerable women whose risks are currently not monitored and are not protected by judicial orders. This is a necessary step in order to engage the necessary means to protect them from extremes forms of violence such as the ones recently witnessed in Spain in the aftermath of the quarantine[22].

[22] https://elpais.com/sociedad/2021-05-21/cinco-asesinatos-machistas-en-cuatro-dias-un-efecto-del-fin-de-las-restricciones.html

5 Conclusion

We estimate that the introduction of the quarantine policies as a response to the pandemic outbreak is responsible for an average increase in IPV reporting of 48% between March 2020 and December 2020, which corresponds to an additional +26,000 calls compared to 2019 levels over the same period. Contrary to previous findings, we find that 67% of these additional calls were made by victims' themselves against 33% for third party calls[23], and that the composition of reported DV in terms of who is reporting the violence changes slightly after lockdown. We find that mobility reduction and previous history of IPV incidence are the main driver of differences in IPV reporting across provinces. This suggests, in turn, that violence is highly sensitive to time spent at home, and that violence might have increased at the intensive margin in the pandemic aftermath.

Differences with previous estimates could steam from the specificity of previous studies, which focused on large and populated cities (usually in the US), differences in the granularity of the data (weekly vs monthly) and the time frame used (several weeks vs several months). They could also steam from differences in the national or local overall system of detection of IPV victims, which is likely to differ significantly across countries and cities.

When looking at the effect of the pandemic outbreak on traditional channels of detection and protection of IPV victims in Spain, we find evidence that reporting through the hotline substituted reporting to the police. We also find that the quarantine caused a "loss" of at least 3,166 protection orders to victims that would, in the absence of the pandemic, have received it. Our results point to the crucial need to detect the most vulnerable women whose risks are currently not being monitored and who are not protected by judicial orders, in order to engage the necessary means to protect them from extremes forms of violence such as the ones recently witnessed in Spain in the aftermath of the quarantine[24].

[23] To obtain the share of additional DV reporting due to help-seeking behavior, we divide the estimated additional calls by victims (specification 5) by the additional total calls.

[24] https://elpais.com/sociedad/2021-05-21/cinco-asesinatos-machistas-en-cuatro-dias-un-efecto-del-fin-de-las-restricciones.html

Bibliography

Arenas-Arroyo, E., Fernandez-Kranz, D. and Nollenberger, N. (2020) 'Can't Leave You Now ! Intimate Partner Violence under Forced Coexistence and Economic Uncertainty', *IZA Discussion Paper*, (13570).

Beland, L.-P. *et al.* (2020) 'COVID-19, Family Stress and Domestic Violence: Remote Work, Isolation and Bargaining Power', *IZA Discussion Papers*, (13332). Available at: https://www.iza.org/publications/dp/13332/covid-19-family-stress-and-domestic-violence-remote-work-isolation-and-bargaining-power.

Campedelli, G. M., Aziani, A. and Favarin, S. (2020) 'Exploring the Effect of 2019-nCoV Containment Policies on Crime: The Case of Los Angeles', pp. 1–49. doi: 10.31219/osf.io/gcpq8.

Ivandic, R., Kirchmaier, T. and Linton, B. (2021) 'The Role of Exposure in Domestic Abuse Victimization: Evidence from the COVID-19 Lockdown'.

Kumar, A. and Nayar, K. R. (2020) 'COVID 19 and its mental health consequences', *Journal of mental health (Abingdon, England)*, 0(0), pp. 1–2. doi: 10.1080/09638237.2020.1757052.

Leslie, E. and Wilson, R. (2020) 'Sheltering in place and domestic violence: Evidence from calls for service during COVID-19', *Journal of Public Economics*, 189, p. 104241. doi: 10.1016/j.jpubeco.2020.104241.

Miller, A., Segal, C. and Spencer, M. (2020) 'Effects of the COVID-19 Pandemic on Domestic Violence in Los Angeles', *NBER Working Paper*. Available at: http://www.nber.org/papers/w28068.pdf.

Perez-Vincent, S. M. *et al.* (2020) 'COVID-19 Lockdowns and Domestic Violence: Evidence from Two Studies in Argentina', *inter- American Development Bank*, pp. 1–48.

Peterman, A. *et al.* (2020) 'Pandemics and Violence Against Women and Children', *Center for Global Development*, (528), p. 43.

Piquero, A. R. *et al.* (2020) 'Staying Home, Staying Safe? A Short-Term Analysis of COVID-19 on Dallas Domestic Violence', *American Journal of Criminal Justice*, pp. 601–635. doi: 10.1007/s12103-020-09531-7.

Ravindran, S. and Shah, M. (2020) 'Unintended consequences of lockdowns: COVID-19 and the shadow pandemic', *NBER Working Paper*.

Appendix

a. Chronology of COVID-19 related measures in Spain

State of emergency and strict quarantine (13th of March to 28th of April): the state of emergency was declared on the 13th of March 2020. Two days later, national lockdown was implemented on the 15th of March 2020 under very strict conditions: mobility was limited to essential needs and restrictions included the closure of bars, restaurants, night clubs, movie theatres and theatres, etc. While most countries maintained authorization to go outside for physical activity or stroll around even at the highest peak of the pandemic (particularly for families with children), in the Spanish case this was not allowed until two months later during the 1st phase of the deconfinement measures.

De-escalation phase (28th April to 20th of June) After nearly two months of lockdown under those strict conditions on April 28th, the government announced the Spanish deconfinement plan organized in four different progressive phases. In the 1st and 2nd phase residents could gather in groups of no more than 10 people (15 in the 2nd), practice physical activity or walk outside at specific hours, attend outside events, go to the library, attend places of cult. Museums, theatres, restaurants, shops were also progressively allowed to open under limited capacity (50%). In the 3rd and last phase, restrictions on the duration and time to go outside were lifted. Some restrictions on group gatherings and capacities were maintained to allow for social distancing).

Progressive deconfinement measures were implemented at different times and under different conditions across the Spanish territory. Between the 11th of May and the 18th of May, 70% of all Spanish provinces entered the 1st deconfinement phase, which allowed limited outside activities and small group gatherings. On the other hand, the most strongly hit autonomous communities (Barcelona, Madrid, and part of the region of Castilla La Mancha) remained in the initial phase of the quarantine longer than the others, until the 25th of May (later for some provinces). Additional protective measures were also taken during the progressive deconfinement phases in Barcelona and Madrid, where mobility to and from the two areas was monitored while bars, restaurants, museums, etc. were allowed to open.

"New normality" (20th June to 25th of October): The state of emergency declared on the 13th of March expired the 21th of June, and Spain entered a "new normality" phase, during which all internal and international borders were opened. Maintained restrictions were limited to regulation of maximum occupancy in shops and masks in public when social distancing was not possible. During this new normality phase, some provinces experience significant increases in the

number of cases and deaths due to COVID-19, which paved the way for local implementation of lockdown in Madrid on the 1st of October.

Re-imposition of State of Emergency (25th October to 9th of May): on the 21st of October Spain passed 1 million COVID-19 cases. 4 days later, on the 25th, the government reimposed a state of emergency across all provinces and introduced a national curfew to counter the resurgence. Local authorities were also given powers to ban travel across different regions and to regulate opening of bars and restaurants. On the 9th of May, the state of emergency came to an end: curfew was lifted in all Spanish provinces, together with restrictions on bars and restaurants. Restrictions regarding occupation capacity were relaxed but maintained.

b. Mobility reduction during lockdown

Description of the mobility database

The Spanish National Institute of Statistics (INE) created a mobility data base using mobile phone tracking information provided by the three main phone operators in Spain with the aim of tracking population movements in the context of the pandemic.

This mobility data base contains daily information on the share of the mobile population at the province and CC.AA level during lockdown between March 16th and June 2020th[25], and during a reference period - the week of the 18th of November 2019. For each considered geographical level, the mobile population is defined as the population which left its residential area for more than 2 hours between 10am and 4pm. Residential areas were defined by the INE based on census delimitations and comprise between 5,000 and 50,000 inhabitants. The largest city in Spain, Madrid, is divided into 150 residential areas of approximately 20,000 inhabitants.

We use this information to define a *mobility reduction indicator* at the province and monthly level, which corresponds to the monthly average in mobility reduction compared to pre-covid (November) levels.

The determinants of effective mobility across provinces

Mobility reduction differences during confinement could steam from pre-pandemic province characteristics (such as demographics, labor market structure, etc.) or from differences in the evolution of the pandemic across the Spanish territory– with mobility being more restricted in the

[25] Daily mobility data in March is given for every other day, and for everyday in the following months.

most hit provinces, either because of a stronger enforcement of the confinement measures by the local police or because of a stricter quarantine compliance by the population due to the fear of being infected/spreading the virus. We formally quantify to what extent mobility reduction reflects other area characteristics by running an OLS regression of the monthly decrease in mobility compared to November 2019, on yearly population density, quarterly unemployment rates, and COVID-19 incidence as measured by the monthly number of new-cases per 100,000 inhabitants. We note that COVID-19 incidence is endogenous to mobility reduction, as mobility reduction affects the diffusion of the virus and as higher incidence of COVID-19 cases is likely to encourage people to stay at home (or policy makers to strengthen social distancing measures). Nonetheless, results displayed in the table suggest that none of the explanatory variables considered explain significant differences between provinces, once province and time fixed effects are controlled for (specification 6).

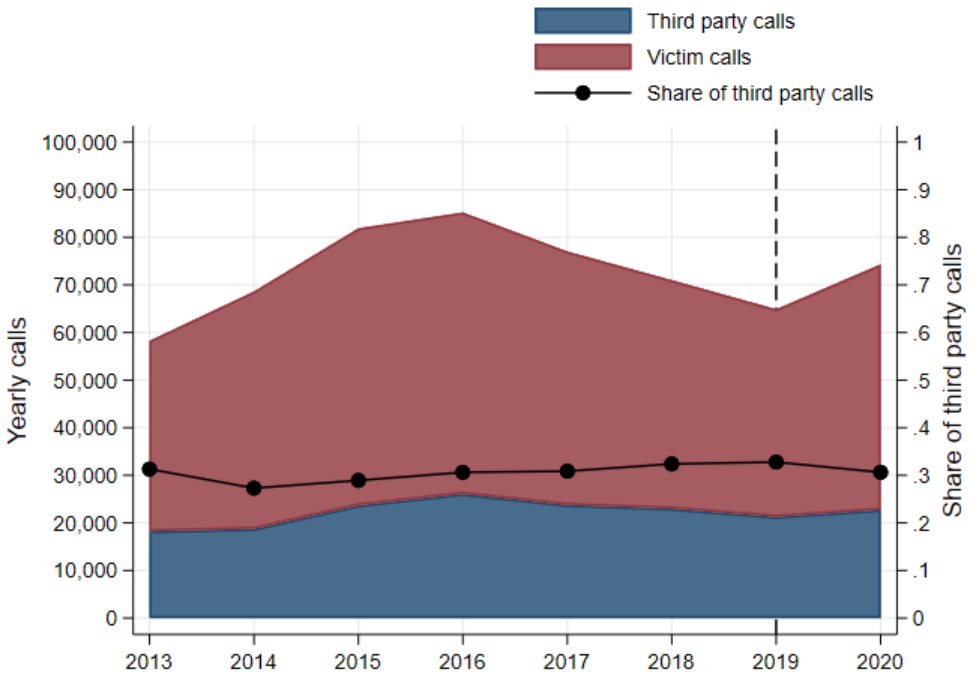
Table A 1. Factors of variation of mobility reduction across provinces between March and June 2020: OLS estimates

	(1) Mobility reduction, density	(2) Mobility reduction, COVID-19 incidence	(3) Mobility reduction, unemployment	(4) Mobility reduction, all controls w/o FE	(5) Mobility reduction, all controls w/ province FE	(6) Mobility reduction, all controls and FE
Population density	0.081			-0.001***	0.289	0.089
	(0.085)			(0.000)	(0.375)	(0.089)
Monthly COVID-19 cases per 100,000		0.001		-0.015***	-0.017***	0.001
		(0.001)		(0.003)	(0.003)	(0.001)
Quaterly Female unemployment rate			-0.073	0.450***	0.148	-0.105
			(0.101)	(0.118)	(0.409)	(0.087)
Quaterly Male unemployment rate			0.109	-0.598**	-1.243**	0.068
			(0.119)	(0.180)	(0.433)	(0.114)
Constant	-27.114	0.000	-0.138	-9.404***	-2.131	-28.915
	(28.578)	(0.386)	(1.993)	(1.239)	(11.309)	(29.496)
Observations	260	260	260	260	260	260
Within R^2	0.953	0.953	0.953	0.259	0.400	0.954
Province FE	Yes	Yes	Yes	No	Yes	Yes
Month and year FE	Yes	Yes	Yes	No	No	Yes

Note: The outcome of interest is the monthly decrease in the share of the mobile population in province p and month m, compared to the share of the mobile population in November 2019 in province p. We regress the monthly decrease in mobility compared to November 2019 on population density, COVID-19 incidence as measured by the monthly number of cases per 100,000 inhabitants, quarterly male and female unemployment rate.
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

c. Tables and figures

Fig A 1. Yearly number of calls according to type of caller over all Spanish provinces, 2013-2020



Source: National Office for gender-based violence

Fig A 2. Yearly number of calls according to type of caller over all Spanish provinces, 2013-2020

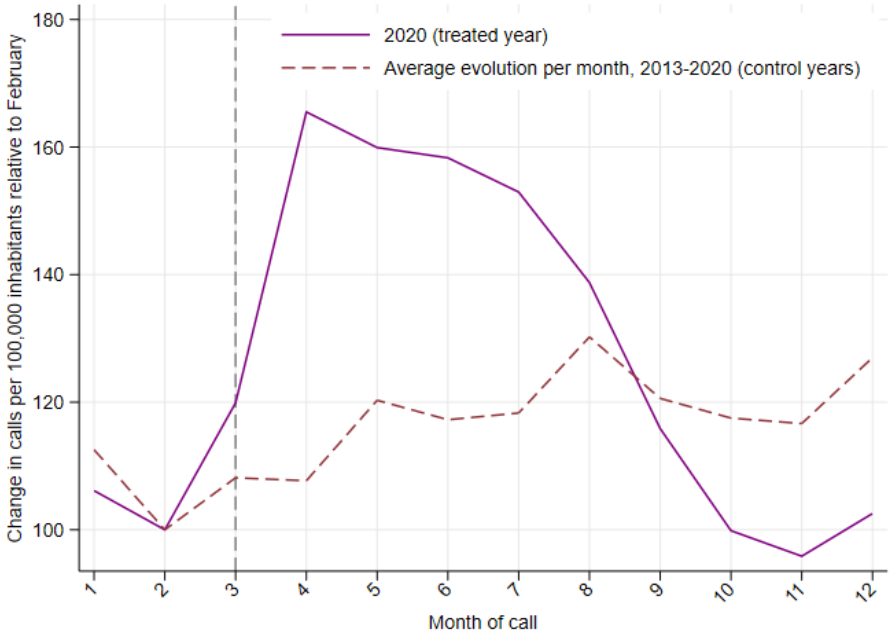


Table A 2. Number of calls in Spain and on average across provinces between March and December 2019

	Total calls in Spain from March to December 2019		Average calls across provinces between March and December 2019	
Caller	*Number of calls*	*Calls per 100 000 inhabitants*	*Number of calls [std in brackets]*	*Calls per 100 000 inhabitants [std in brackets]*
Total	55,233	117.7	1062.2 [1855.4]	99.8 [30.21]
Victim	37,117	79.1	713.8 [1279.8]	66.7 [22.08]
Third party	18,116	38.6	348.3 [578.9]	33.1 [10.09]

Table A 3. Impact of lockdown introduction on <u>total</u> domestic violence reporting through emergency hotline – difference-in-differences estimates: log models with controls

	(1) Log of calls, controlling for density	(2) Log of calls, controlling for unemp rate	(3) Log of calls, controlling for male and female unemp rate	(4) Log of calls, all controls
$\boldsymbol{D_{y=2020}D_{m\geq 3}}$	**0.390*** **	**0.389*** **	**0.388*** **	**0.389*** **
	(0.026)	**(0.026)**	**(0.027)**	**(0.027)**
$\boldsymbol{D_{m\geq 3}}$	0.040+	0.037+	0.038+	0.039+
	(0.023)	(0.021)	(0.022)	(0.021)
$\boldsymbol{D_{y=2020}}$	0.035	0.017	0.021	0.027
	(0.025)	(0.038)	(0.040)	(0.041)
Population density in 2019	-0.001***			-0.001*
	(0.000)			(0.000)
Quarterly unemp. rate		-0.002		
		(0.003)		
Quarterly female unemp. rate			-0.002	-0.001
			(0.002)	(0.003)
Quaterly male unemp. rate			0.000	0.000
			(0.003)	(0.003)
Constant	3.708***	3.470***	3.464***	3.699***
	(0.330)	(0.309)	(0.313)	(0.336)
Observations	4990	4990	4990	4990
R^2	0.216	0.215	0.215	0.216
Province FE	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes

<u>Note</u>: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March, and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years.

<u>Significance level</u>: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A 4. Impact of lockdown introduction on victims' domestic violence reporting through emergency hotline – difference-in-differences estimates: log models with controls

	(1) Log of calls, controlling for density	(2) Log of calls, controlling for unemp rate	(3) Log of calls, controlling for male and female unemp.	(4) Log of calls, all controls
$\boldsymbol{D_{y=2020}D_{m\geq 3}}$	**0.380*** **	**0.381*** **	**0.377*** **	**0.378*** **
	(0.030)	**(0.030)**	**(0.031)**	**(0.031)**
$\boldsymbol{D_{m\geq 3}}$	0.035	0.038+	0.041+	0.042*
	(0.023)	(0.021)	(0.021)	(0.021)
$\boldsymbol{D_{y=2020}}$	0.062*	0.084+	0.095+	0.103*
	(0.030)	(0.045)	(0.047)	(0.048)
Population density in 2019	-0.002***			-0.002*
	(0.000)			(0.001)
Quarterly unemp. rate		0.002		
		(0.003)		
Quarterly female unemp. rate			-0.003	-0.002
			(0.002)	(0.003)
Quaterly male unemp. rate			0.005	0.005
			(0.003)	(0.003)
Constant	3.406***	2.999***	2.983***	3.313***
	(0.339)	(0.308)	(0.320)	(0.355)
Observations	4984	4984	4984	4984
R^2	0.177	0.177	0.178	0.178
Province FE	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March, and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years.

Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A 5. Impact of lockdown introduction on third parties' domestic violence reporting through emergency hotline – difference-in-differences estimates: log models with controls

	(1) Log of calls, controlling for density	(2) Log of calls, controlling for unemp rate	(3) Log of calls, controlling for male and female unemp	(4) Log of calls, all controls
$\boldsymbol{D_{y=2020}D_{m\geq 3}}$	**0.414*** **	**0.381*** **	**0.377*** **	**0.378*** **
	(0.035)	**(0.030)**	**(0.031)**	**(0.031)**
$\boldsymbol{D_{m\geq 3}}$	0.028	0.038+	0.041+	0.042*
	(0.029)	(0.021)	(0.021)	(0.021)
$\boldsymbol{D_{y=2020}}$	-0.029	0.084+	0.095+	0.103*
	(0.029)	(0.045)	(0.047)	(0.048)
Population density in 2019	-0.002***			-0.002*
	(0.000)			(0.001)
Quarterly unemp. rate		0.002		
		(0.003)		
Quarterly female unemp. rate			-0.003	-0.002
			(0.002)	(0.003)
Quaterly male unemp. rate			0.005	0.005
			(0.003)	(0.003)
Constant	2.608***	2.999***	2.983***	3.313***
	(0.387)	(0.308)	(0.320)	(0.355)
Observations	4949	4984	4984	4984
R^2	0.133	0.177	0.178	0.178
Province FE	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March, and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years.

Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Fig A 3. Impact of lockdown introduction on <u>total</u> domestic violence reporting through emergency hotline – event study estimates: log model

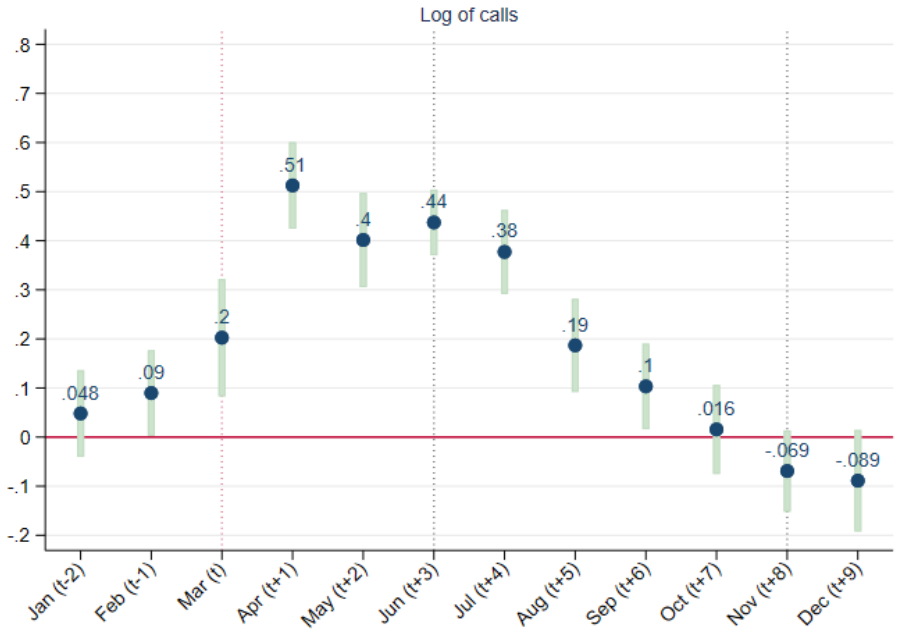


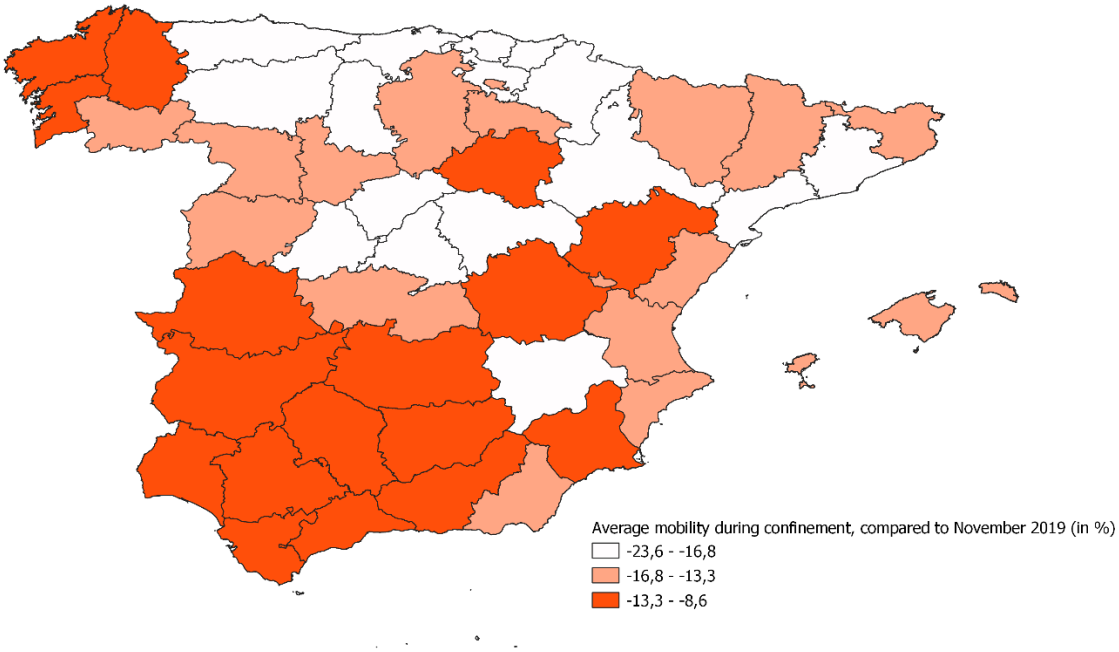
Table A 6. Share of the mobile population and mobility reduction in Spain, per month

Month	Average share of the mobile population across provinces	Average decrease of the mobile population across provinces
November (reference month) 2019	26.9%	0.0pp
March 2020	9.5%	-17.4pp
April 2020	8.6%	-18.3pp
May 2020	12.6%	-14.3pp
June 2020	16.6%	-10.3pp

Source: INE, Statistical information for the analysis of the COVID-19 crisis' impact. Mobility data.

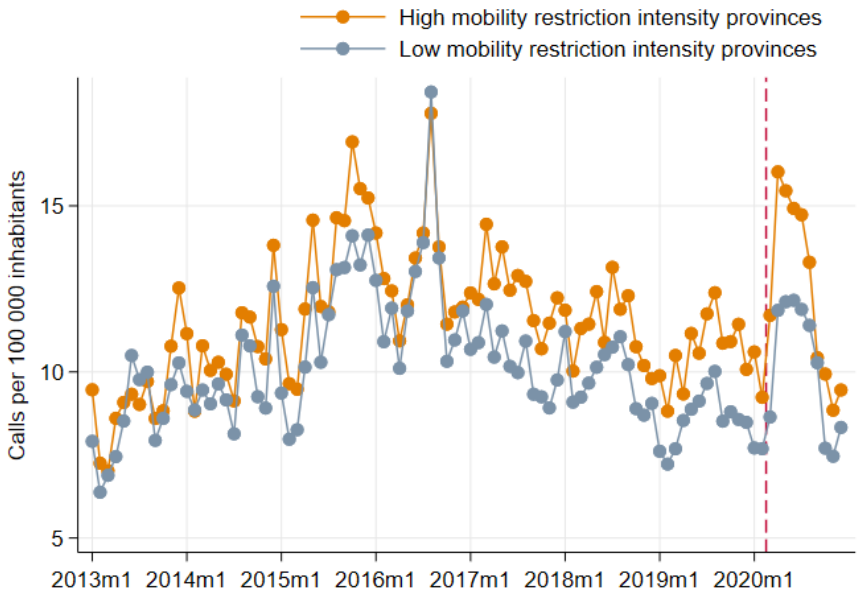
Note: Mobility reduction is the difference between the share of the mobile population in the considered month, with the share of the mobile population in the reference month (November)

Fig A 4. Average decrease in the share of the mobile population between the 15th of March 2020 and the 20th of June, compared to November 2019



Source: INE

Fig A 5. Monthly calls per 100,000 inhabitants between 2013 and 2020, in high and low mobility reduction provinces



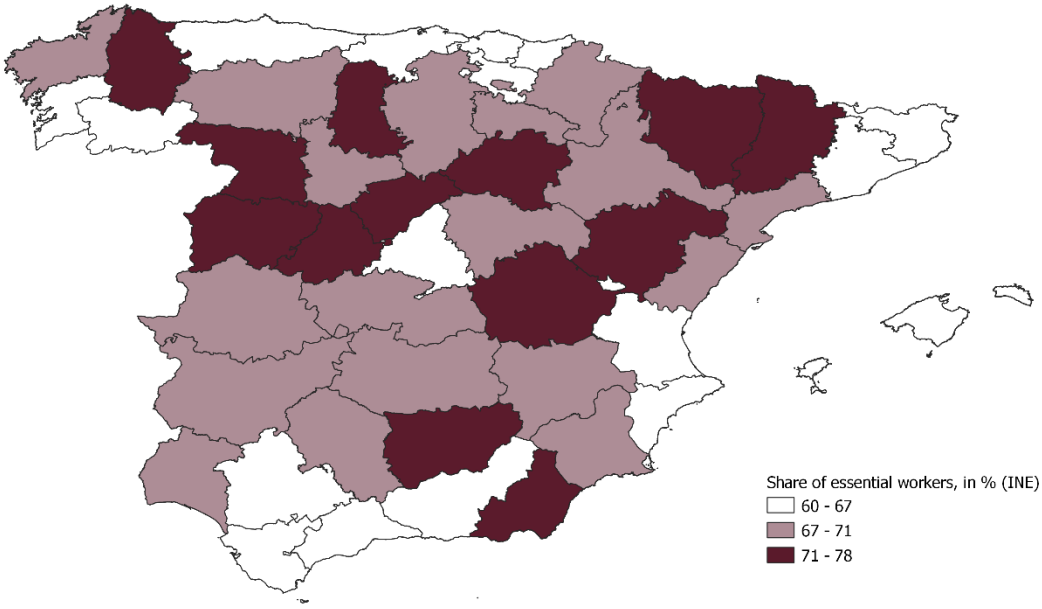
Source: INE

Fig A 6. Police reports for IPV and protection orders per 100,000 inhabitants in Spain, between 2017 and 2020

Source: INE

Note: for each considered measure of IPV incidence (police reports or protection orders), high vs low pre-pandemic IPV incidence categories are defined based on the average value of IPV incidence per 100,000 inhabitants between 2015 and 2018. Provinces with average values above (resp. below or equal to) the median are considered as high (resp. low) pre-pandemic IPV incidence provinces.

Fig A 7. Share of essential workers by province, as per Royal Decree 10/2020 (INE)



Source: INE

Table A 7. Effect of lockdown introduction on quarterly police complaints for IPV and protection orders, DiD estimates

	(1) Police reports for IPV (log)	(2) Police reports for IPV filed by victims (log)	(3) Protection orders (log)	(4) Police reports for IPV per 100,000	(5) Police reports for IPV *filed by victims* per 100,000	(6) Protection orders per 100,000
$\boldsymbol{D_{y=2020}D_{q\geq 2}}$	**0.015**	**0.037**	**-0.114****	**0.750**	**1.648**	**-2.249****
	(0.025)	(0.034)	(0.033)	(1.470)	(1.524)	(0.666)
$\boldsymbol{D_{q\geq 2}}$	0.001	0.004	0.007	2.128*	1.195	0.681**
	(0.009)	(0.016)	(0.013)	(0.830)	(1.006)	(0.236)
$\boldsymbol{D_{y=2020}}$	0.076+	0.060	0.027	13.288***	9.983***	3.791**
	(0.039)	(0.050)	(0.072)	(2.057)	(2.100)	(1.227)
Constant	5.827***	5.725***	4.814***	54.954***	41.047***	14.965***
	(0.171)	(0.214)	(0.352)	(1.255)	(1.211)	(0.707)
Observations	2496	2496	2496	1664	1664	1664
R^2	0.301	0.132	0.091	0.360	0.265	0.133
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter and year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{q\geq 2}$ is a dummy variable which is equal to 0 from January to March, and 1 for the 2nd 3rd and 4th quarter of year. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction from the 2nd quarter of year onwards compared to previous years.

Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001